

# Constraint Inventory for the Exact Certification of the Level Profile $\zeta(a) = (5a - 4)/4$

Companion documentation to Appendix A  
of “A One-Hypothesis Reduction for Primes in  $[4n^2 - n, 4n^2 + n]$ ”

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## Abstract

This note is the full constraint inventory referred to in Appendix A of the paper referenced above (henceforth [P]). It derives every linear inequality appearing in the certification script `fm_exact.py` from the published theorem that inequality encodes; states the sign, negation, and boundary conventions needed to read the code’s coefficient tuples; records the three places where the encoding is a deliberate *relaxation* of the true tool, and proves that the certification remains sound there by a sandwich against the Coverage Lemma of [P]; and cross-references every atom family to the code identifiers and to the corresponding statements of [P]. This is designed to be an audit trail, not new mathematics.

## 1 Purpose and provenance

The script `fm_exact.py` certifies, in exact rational arithmetic, the level profile

$$\zeta(\gamma) = \frac{6 - 5\gamma}{4} \quad (\gamma \in [1, 6/5]),$$

in both directions: for  $\beta < \zeta(\gamma)$  every estimation block admits an admissible configuration (achievability — proved by hand in [P], Coverage Lemma, and re-verified here as a byproduct), and at  $\beta = \zeta(\gamma)$  a block exists that no encoded tool covers (optimality — [P], Optimality Proposition). The certification is a finite computation because every tool’s admissibility region is a conjunction of inequalities that are *linear* in the block coordinates at fixed  $\gamma$ . This document is the dictionary between those inequalities, the code, and the literature.

## 2 Conventions

*Convention 2.1* (coordinates and exponents). A slice is fixed by  $\gamma \in [1, 6/5]$  (prime block  $p \sim N^\gamma$ ; equivalently  $a = 2 - \gamma$ , cofactor  $q \asymp N^a$ ). The unknowns are  $x = (\mu, \eta, \beta)$ : the Vaughan-piece exponent, the dyadic frequency exponent, and the sieve-level exponent. All exponents are with respect to base  $N = 2n$ ;  $x_{\text{arith}} := 4n^2 \asymp N^2$ . We write  $\Sigma := \beta + \gamma$  and  $\xi := \eta + 2 - \Sigma$  (phase-size exponent). The admissibility requirement for a block, [P] eq. (trilinear

requirement), is that the estimating bound have exponent at most  $\Sigma$  (i.e. relative saving  $N^\eta$  against the trivial size  $N^{\eta+\Sigma}$ ), with an  $x^{-c\varepsilon}$  margin suppressed throughout this document: all atoms are stated with  $\varepsilon = 0$  and *closed* inequalities. Facet equalities are therefore boundary points of both the pass and fail regions; this is intentional (see Convention 2.3).

*Convention 2.2* (atom format in the code). In `failure_atoms()` a constraint is the tuple  $C(a_\mu, a_\eta, a_\beta, c_0, c_\gamma)$ , meaning

$$a_\mu \mu + a_\eta \eta + a_\beta \beta \leq c_0 + c_\gamma \gamma,$$

all entries rational. Each *tool* passes on a conjunction of pass conditions; the code stores, per tool, the list of *negated* pass conditions (“fail atoms”), one of which must hold for the tool to fail at a point. A *failure witness* is a point where every tool fails; distributing the disjunctions gives the product over tools of the fail-atom counts:

$$\underbrace{3}_{\text{FI0}} \times \underbrace{3}_{\text{FI1}} \times \underbrace{3}_{\text{FI2}} \times \underbrace{2}_{\text{RS0}} \times \underbrace{2}_{\text{RS1}} \times \underbrace{2}_{\text{RS2}} \times \underbrace{1}_{\text{RS3}} \times \underbrace{1}_{\text{ELEM}} = 216$$

candidate polytopes per piece type (ELEM is present only for type I, where its single fail option is the conjunction of its three negated clauses; type II has 216 combinations without the final factor). Each polytope comprises the chosen fail atoms plus the region constraints of Section 3 ( $\approx 14$  inequalities in 3 unknowns); the minimal  $\beta$  over it is found by exhaustive vertex enumeration ( $\binom{14}{3} = 364$  candidate bases,  $3 \times 3$  systems solved by Cramer’s rule over  $\mathbb{Q}$  in `solve3()`), and  $\zeta(\gamma)$  is the minimum over all polytopes (`min_beta()`).

*Convention 2.3* (boundaries and monotonicity). Because all fail atoms are closed, the computed  $\zeta(\gamma)$  is the infimum of levels admitting a witness, and the witness printed at  $\beta = \zeta(\gamma)$  sits on the facet with equalities active. Failure is not a priori monotone in  $\beta$  (raising  $\beta$  widens the frequency range and tightens windows, but also widens the reachable grouping intervals), so the script’s output is, by itself, only the statement “the *first* failure occurs at  $\beta = \zeta(\gamma)$ .” That everything strictly below the facet passes is the Coverage Lemma of [P], proved by hand; the two artifacts meet at the facet. See also Section 7.

### 3 The block model and the region $R$

Per [P] (Decomposition and grouping), a block carries variables  $(h, e_1, e_2, m_v, k)$  with exponents  $(\eta, \beta - s, s, \mu, \gamma - \mu)$ ,  $s \in [0, \beta]$  free. Vaughan pieces: type I has  $\mu \in [0, 2\gamma/3]$  with  $k$  smooth; type II has  $\mu \in [\gamma/3, \gamma/2]$ , both coefficients arbitrary. The frequency range from the pipeline lemma is  $0 \leq \eta \leq \Sigma - 1$  (blocks with  $\Sigma \leq 1$  require nothing). The region constraints in `region()`:

| code tuple                            | inequality                     | role            | piece type |
|---------------------------------------|--------------------------------|-----------------|------------|
| $C(-1, 0, 0, 0, 0)$                   | $\mu \geq 0$                   | piece range     | I          |
| $C(1, 0, 0, 0, 2/3)$                  | $\mu \leq 2\gamma/3$           | piece range     | I          |
| $C(-1, 0, 0, 0, -1/3)$                | $\mu \geq \gamma/3$            | piece range     | II         |
| $C(1, 0, 0, 0, 1/2)$                  | $\mu \leq \gamma/2$            | piece range     | II         |
| $C(0, -1, 0, 0, 0)$                   | $\eta \geq 0$                  | frequency range | both       |
| $C(0, 1, -1, -1, 1)$                  | $\eta \leq \beta + \gamma - 1$ | frequency range | both       |
| $C(0, 0, -1, 0, 0), C(0, 0, 1, 1, 0)$ | $0 \leq \beta \leq 1$          | search box      | both       |

## 4 Tool family RS: the Robert–Sargos theorem

### 4.1 Statement and specialization

Robert–Sargos [3, Theorem 1]: for exponents with  $\alpha(\alpha-1)\beta_0\gamma_0 \neq 0$ , dyadic ranges  $H, N_1, M_1$ , phase size  $X \geq 1$ , and arbitrary coefficients  $|a(h, \nu)| \leq 1$ ,  $|b(w)| \leq 1$ ,

$$\sum_{h \sim H} \sum_{\nu \sim N_1} a(h, \nu) \sum_{w \sim M_1} b(w) e\left(X \frac{h^{\beta_0} \nu^{\gamma_0} w^{\alpha}}{H^{\beta_0} N_1^{\gamma_0} M_1^{\alpha}}\right) \ll_{\varepsilon} (HN_1M_1)^{1+\varepsilon} \left\{ \left(\frac{X}{HN_1M_1^2}\right)^{1/4} + (HN_1)^{-1/4} + M_1^{-1/2} + X^{-1/2} \right\}.$$

We use  $(\beta_0, \gamma_0, \alpha) = (1, -1, -1)$ , so  $\alpha(\alpha-1)\beta_0\gamma_0 = -2 \neq 0$ ; the standalone slot  $w$  has exponent  $m_1$ , the joint slot  $(h, \nu)$  has exponents  $(\eta, \Sigma - m_1)$ , and  $X = N^{\xi}$  with  $\xi \geq 2 - \Sigma \geq 3/4 > 0$ .

### 4.2 Window derivation

The requirement is  $(\eta + \Sigma) + \max_i t_i \leq \Sigma$ , i.e.  $\max_i t_i \leq -\eta$ , where in exponents, with  $n_1 = \Sigma - m_1$  and  $\xi - \eta = 2 - \Sigma$ :

$$t_1 = \frac{\xi - \eta - n_1 - 2m_1}{4} = \frac{2 - 2\Sigma - m_1}{4} \Rightarrow m_1 \geq 4\eta + 2 - 2\Sigma \quad (\text{A})$$

$$t_2 = -\frac{\eta + n_1}{4} \Rightarrow m_1 \leq \Sigma - 3\eta \quad (\text{B})$$

$$t_3 = -\frac{m_1}{2} \Rightarrow m_1 \geq 2\eta \quad (\text{C})$$

$$t_4 = -\frac{\xi}{2} \Rightarrow \eta \leq 2 - \Sigma \quad (\text{D})$$

Subsumptions used by the encoding: (A)  $\Leftarrow$  (C) since  $4\eta + 2 - 2\Sigma \leq 2\eta \iff \eta \leq \Sigma - 1$ , which is a region constraint; (D) is implied on the relevant part of the region (see Remark 4.1). The effective window is  $m_1 \in [2\eta, \Sigma - 3\eta]$ , matching [P] (Admissibility window Lemma), whose window is nonempty iff  $5\eta \leq \Sigma$ .

### 4.3 The four grouping families and their code atoms

The reachable standalone exponents are  $m_1 \in \{s, \mu + s, \gamma - \mu + s, \gamma + s\}$ ,  $s \in [0, \beta]$  ([P], available-set display; the fourth family is retained here because the RS window's upper cap  $\Sigma - 3\eta$  can exceed  $\gamma$  at small  $\eta$ ). Taking in each family the choice  $m_1 = \max(\text{base}, 2\eta)$  where possible, the pass conditions and their negations:

| tool                             | pass conditions                                                     | fail atoms (code)              | fail inequality                                                     |
|----------------------------------|---------------------------------------------------------------------|--------------------------------|---------------------------------------------------------------------|
| RS0 ( $m_1 = s$ )                | $2\eta \leq \beta; 5\eta \leq \Sigma$                               | $\mathbf{C}(0, -2, 1, 0, 0)$   | $\beta \leq 2\eta$                                                  |
|                                  |                                                                     | $\mathbf{C}(0, -5, 1, 0, -1)$  | $\beta \leq 5\eta - \gamma$ (i.e. $5\eta \geq \Sigma$ )             |
| RS1 ( $m_1 = \mu + s$ )          | $\mu + \beta \geq 2\eta; \mu \leq \Sigma - 3\eta$                   | $\mathbf{C}(1, -2, 1, 0, 0)$   | $\mu + \beta \leq 2\eta$                                            |
|                                  |                                                                     | $\mathbf{C}(-1, -3, 1, 0, -1)$ | $\mu \geq \Sigma - 3\eta$                                           |
| RS2 ( $m_1 = \gamma - \mu + s$ ) | $\gamma - \mu + \beta \geq 2\eta; \gamma - \mu \leq \Sigma - 3\eta$ | $\mathbf{C}(-1, -2, 1, 0, -1)$ | $\gamma - \mu + \beta \leq 2\eta$                                   |
|                                  |                                                                     | $\mathbf{C}(1, -3, 1, 0, 0)$   | $\mu + \beta \leq 3\eta$ (i.e. $\gamma - \mu \geq \Sigma - 3\eta$ ) |
| RS3 ( $m_1 = \gamma + s$ )       | $3\eta \leq \beta$ (i.e. $\gamma \leq \Sigma - 3\eta$ )             | $\mathbf{C}(0, -3, 1, 0, 0)$   | $\beta \leq 3\eta$                                                  |

*Remark 4.1* (documented relaxations in the RS encoding). Three omissions make the encoded tools slightly *stronger* than the true ones on part of the search box: (i) **RS1** and **RS2** omit the window-nonemptiness condition  $5\eta \leq \Sigma$  (which **RS0** includes); (ii) **RS3** omits the lower-edge

condition  $\gamma + s \geq 2\eta$ , vacuous only when  $2\eta \leq \gamma$ , i.e. when  $\Sigma \leq (\gamma + 2)/2$ ; (iii) condition (D) of Subsection 4.2 is omitted everywhere, valid only when  $\eta \leq 2 - \Sigma$ , i.e. when  $\Sigma \leq 3/2$  given  $\eta \leq \Sigma - 1$ . All three omitted conditions *hold automatically* on  $\{\Sigma \leq 5/4, \eta \leq \Sigma - 1\}$ , which contains the entire facet region; outside it they are genuine relaxations. Soundness is unaffected: see Section 7.

## 5 Tool family ELEM: elementary bounds (type I only)

For a type I block the smooth variable  $k \sim K = N^\kappa$ ,  $\kappa = \gamma - \mu$ , is summed with the phase  $f(k) = hz'/(ck)$ ,  $c = e_1 e_2 m_v \sim N^\chi$ ,  $\chi = \beta + \mu$ , all other variables in absolute value;  $|f'| \asymp \Lambda = N^\lambda$ ,  $\lambda = \eta + 2 - \chi - 2\kappa$ . Three clauses, whose disjunction is the pass condition; the tool fails only if all three fail, which is the single ELEM conjunction in the code.

(E1) *Kuzmin–Landau branch* ( $\lambda < 0$ ): by Kuzmin–Landau (e.g. [2, Theorem 2.1], after dyadic subdivision so that  $|f'| \leq \frac{1}{2}$ ),  $\sum_k e(f(k)) \ll \Lambda^{-1}$ ; total exponent  $\eta + \chi - \lambda = 2(\chi + \kappa) - 2 = 2\Sigma - 2 \leq \Sigma$  for  $\Sigma \leq 2$ : *always admissible in region*. Pass clause:  $\lambda < 0 \iff \eta \leq \beta - \mu + 2\gamma - 2$ .

Fail atom  $\mathbf{C}(-1, -1, 1, 2, -2)$ :  $\eta \geq \beta - \mu + 2\gamma - 2$ .

(E2) *Second-derivative test / exponent pair* ( $\frac{1}{2}, \frac{1}{2}$ ) [2, Theorem 2.2]:  $\sum_k \ll (\Lambda K)^{1/2} + (K/\Lambda)^{1/2}$ . The first term needs  $\eta + \chi + \frac{\lambda + \kappa}{2} \leq \Sigma$ , which simplifies to  $3\eta + 2\mu \leq \beta + 3\gamma - 2$ ; the companion term needs  $\eta + \chi + \frac{\kappa - \lambda}{2} = \frac{\eta + 3\Sigma - 2}{2} \leq \Sigma$ , i.e.  $\eta \leq 2 - \Sigma$ , automatic on the facet region.

Fail atom  $\mathbf{C}(-2, -3, 1, 2, -3)$ :  $3\eta + 2\mu \geq \beta + 3\gamma - 2$ .

(E3) *Exponent pair* ( $\frac{1}{6}, \frac{2}{3}$ ) [2, §3]:  $\sum_k \ll \Lambda^{1/6} K^{2/3} (+ \text{admissible companions as in (E1), (E2)})$ ; the main term needs  $\eta + \chi + \frac{\lambda}{6} + \frac{2\kappa}{3} = \frac{7\eta + 5\beta + 3\mu + 2\gamma + 2}{6} \leq \Sigma$ , which simplifies to  $7\eta + 3\mu \leq \beta + 4\gamma - 2$ .

Fail atom  $\mathbf{C}(-3, -7, 1, 2, -4)$ :  $7\eta + 3\mu \geq \beta + 4\gamma - 2$ .

*Remark 5.1* (P). uses only (E1) and (E2) in the Coverage Lemma; (E3) is carried in the certification solely so that the Optimality Proposition quantifies over the larger toolkit. The certified optimum is identical with or without (E3) — one of the two byproducts recorded in [P], Appendix A.

## 6 Tool family FI: Fouvry–Iwaniec Theorem 6

This family is *not used* in the proofs of [P]; it is encoded so that the Optimality Proposition’s claim — the profile is unchanged when [1, Theorem 6] is adjoined — is machine-checked rather than asserted. Since the derivation appears nowhere in [P], it is recorded here in full.

### 6.1 Statement and exponent form

[1, Theorem 6]: for  $\alpha \neq 0, 1$ , an additive character  $\chi(h)$ , and arbitrary  $|\varphi_m|, |\psi_n| \leq 1$ ,

$$S_{\chi\varphi\psi}(H, M, N) \ll (HMN)^{1/2} \left[ (H + N)^{1/2} (x_6^{1/8} H^{-1/6} M^{1/12} N^{1/6} + x_6^{1/8} H^{-1/8} N^{3/8} + N^{1/2} + N^{1/4} M^{1/8}) x_6^{1/8} + M^{1/2} + x_6^{-1/4} M^{1/2} N \right] \mathcal{L}^4,$$

where  $x_6$  is the phase size. With slot exponents  $(m_0, n_0)$ ,  $m_0 + n_0 = \Sigma$ , frequency  $\eta$ , and  $x_6 = N^\xi$ , the requirement (bound  $\leq N^\Sigma$ ) yields, after the prefactor  $(\eta + \Sigma)/2$ :

- the  $M^{1/2}$  term (T5):  $\frac{\eta+\Sigma}{2} + \frac{m_0}{2} \leq \Sigma \iff \eta \leq n_0$  — a *lower* window edge, which also fixes  $(H + N)^{1/2} \mapsto N^{n_0/2}$  in the remaining terms;
- five upper caps on  $n_0$ , derived by direct substitution ( $\xi = \eta + 2 - \Sigma$ ,  $m_0 = \Sigma - n_0$ ):

$$\begin{aligned}
\text{T1 } (x_6^{1/4} H^{-1/6} M^{1/12} N^{1/6}.): \quad & 7n_0 \leq 8\Sigma - 7\eta - 6 \Rightarrow n_0 \leq \text{cap}_1 = \frac{8\Sigma - 7\eta - 6}{7} \\
\text{T2 } (x_6^{1/4} H^{-1/8} N^{3/8}.): \quad & 7n_0 \leq 6\Sigma - 5\eta - 4 \Rightarrow n_0 \leq \text{cap}_2 = \frac{6\Sigma - 5\eta - 4}{7} \\
\text{T3 } (x_6^{1/8} N^{1/2}.): \quad & 8n_0 \leq 5\Sigma - 5\eta - 2 \Rightarrow n_0 \leq \text{cap}_3 = \frac{5\Sigma - 5\eta - 2}{8} \\
\text{T4 } (x_6^{1/8} N^{1/4} M^{1/8}.): \quad & 5n_0 \leq 4\Sigma - 5\eta - 2 \Rightarrow n_0 \leq \text{cap}_4 = \frac{4\Sigma - 5\eta - 2}{5} \\
\text{T6 } (x_6^{-1/4} M^{1/2} N.): \quad & 2n_0 \leq 2 - \Sigma - \eta \Rightarrow n_0 \leq \text{cap}_6 = \frac{2 - \Sigma - \eta}{2}
\end{aligned}$$

(Sample derivation, T2: multiply the requirement  $\frac{\eta+\Sigma}{2} + \frac{n_0}{2} + \frac{\xi}{4} - \frac{\eta}{8} + \frac{3n_0}{8} \leq \Sigma$  by 8 and substitute  $\xi$ :  $5\eta + 7n_0 + 2\Sigma + 4 \leq 8\Sigma$ . The others are identical in kind.)

## 6.2 Pruning lemma

**Lemma 6.1.** *On  $\{\eta \leq \Sigma - 1, \Sigma \leq 4/3\}$ :  $\text{cap}_2 \leq \text{cap}_1$ ,  $\text{cap}_2 \leq \text{cap}_3$ , and  $\text{cap}_2 \leq \text{cap}_4$ . Hence the FI window is  $[\eta, \min(\text{cap}_6, \text{cap}_2)]$ .*

*Proof.*  $\text{cap}_1 - \text{cap}_2 = \frac{2\Sigma - 2\eta - 2}{7} \geq 0$  iff  $\eta \leq \Sigma - 1$ .  $\text{cap}_2 \leq \text{cap}_4$  iff  $5(6\Sigma - 5\eta - 4) \leq 7(4\Sigma - 5\eta - 2)$  iff  $\Sigma + 5\eta \leq 3$ ; with  $\eta \leq \Sigma - 1$  this follows from  $6\Sigma \leq 8$ , i.e.  $\Sigma \leq 4/3$ .  $\text{cap}_2 \leq \text{cap}_3$  iff  $8(6\Sigma - 5\eta - 4) \leq 7(5\Sigma - 5\eta - 2)$  iff  $13\Sigma \leq 5\eta + 18$ , implied by  $\Sigma \leq 4/3 < 18/13$ .  $\square$

## 6.3 Grouping families and code atoms

The reachable  $n_0$  families are  $\{s, \mu + s, \gamma - \mu + s\}$ ,  $s \in [0, \beta]$ ; the fourth family  $n_0 = \gamma + s$  is *provably dead* for this tool, since  $\gamma \geq 1$  while  $\text{cap}_6 \leq \frac{2-\Sigma}{2} \leq \frac{1}{2}$ , and is therefore not encoded. In each family we take  $n_0$  at the base value (lower edge condition base  $+\beta \geq \eta$ ):

| tool | pass conditions                                                                                        | fail atoms (code)      | fail inequality                                                 |
|------|--------------------------------------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------|
| FI0  | $\eta \leq \beta$ ; $\eta \leq \text{cap}_6$ ; $\eta \leq \text{cap}_2$                                | $C(0, -1, 1, 0, 0)$    | $\eta \geq \beta$                                               |
|      |                                                                                                        | $C(0, -3, -1, -2, 1)$  | $3\eta + \beta \geq 2 - \gamma$ ( $\eta \geq \text{cap}_6$ )    |
|      |                                                                                                        | $C(0, -12, 6, 4, -6)$  | $12\eta - 6\beta \geq 6\gamma - 4$ ( $\eta \geq \text{cap}_2$ ) |
| FI1  | $\mu + \beta \geq \eta$ ; $\mu \leq \text{cap}_6$ ; $\mu \leq \text{cap}_2$                            | $C(1, -1, 1, 0, 0)$    | $\mu + \beta \leq \eta$                                         |
|      |                                                                                                        | $C(-2, -1, -1, -2, 1)$ | $2\mu + \eta + \beta \geq 2 - \gamma$                           |
|      |                                                                                                        | $C(-7, -5, 6, 4, -6)$  | $7\mu + 5\eta - 6\beta \geq 6\gamma - 4$                        |
| FI2  | $\gamma - \mu + \beta \geq \eta$ ; $\gamma - \mu \leq \text{cap}_6$ ; $\gamma - \mu \leq \text{cap}_2$ | $C(-1, -1, 1, 0, -1)$  | $\gamma - \mu + \beta \leq \eta$                                |
|      |                                                                                                        | $C(2, -1, -1, -2, 3)$  | $2\mu - \eta - \beta \leq 3\gamma - 2$                          |
|      |                                                                                                        | $C(7, -5, 6, 4, 1)$    | $7\mu - 5\eta + 6\beta \leq \gamma + 4$                         |

(The last two FI2 rows are the sign-flipped forms of  $\gamma - \mu \geq \text{cap}_6$  and  $\gamma - \mu \geq \text{cap}_2$ .) The pruning is a further relaxation outside  $\Sigma \leq 4/3$ , handled by the same soundness argument below. The trivial-size lower edge  $\eta \leq n_0$  is encoded as the first pass condition in each family.

## 7 Soundness of the relaxed encoding

Let  $\zeta_{\text{code}}(\gamma)$  be the script's output and  $\zeta_{\text{true}}(\gamma)$  the first-failure level of the *exactly* encoded toolkit. Every deviation documented in Remark 4.1 and Subsection 6.2 makes an encoded tool pass on a superset of its true pass region; failure witnesses are therefore harder to find, so  $\zeta_{\text{code}} \geq \zeta_{\text{true}}$ . In the other direction, the Coverage Lemma of [P] proves by hand — using only the exactly-valid parts of the encoding, all active inside  $\{\Sigma \leq 5/4\}$  where every omitted condition holds — that all blocks pass for  $\beta < (6 - 5\gamma)/4$ , so  $\zeta_{\text{true}} \geq (6 - 5\gamma)/4$ . Since the script returns  $\zeta_{\text{code}} = (6 - 5\gamma)/4$  exactly at every tested rational  $\gamma$ , the sandwich closes:

$$\frac{6 - 5\gamma}{4} = \zeta_{\text{code}} \geq \zeta_{\text{true}} \geq \frac{6 - 5\gamma}{4}.$$

Finally, the printed witnesses are legitimate witnesses for the *true* system: at  $(\mu, \eta) = (\gamma/2, \Sigma - 1)$  with  $4\beta + 5\gamma = 6$  one checks directly that every omitted condition ( $5\eta \leq \Sigma$ ,  $\eta \leq 2 - \Sigma$ ,  $2\eta \leq \gamma$ ,  $\Sigma \leq 4/3$ ) is satisfied, so no relaxed tool was wrongly credited there.

## 8 Cross-reference table

| code identifier      | this document  | item in [P]                                                                                                 |
|----------------------|----------------|-------------------------------------------------------------------------------------------------------------|
| RS0-RS3              | §4             | §5.4 Admissibility-window Lemma (5.3); §5.3 available-set display (10)                                      |
| ELEM clause (E1)     | §5             | §5.5 elementary Lemma (5.4), Kuzmin–Landau term $\Lambda^{-1}$                                              |
| ELEM clause (E2)     | §5             | §5.5 elementary Lemma (5.4), exponent pair $(\frac{1}{2}, \frac{1}{2})$ ; used in §5.6 Coverage Lemma (5.5) |
| ELEM clause (E3)     | §5             | §5.6 Optimality Prop. (5.6); Appendix A (certification only)                                                |
| FI0-FI2              | §6             | §5.6 Optimality Prop. (5.6); Appendix A byproducts (certification only)                                     |
| region()             | §3             | §5.3 Decomposition and grouping; §5.1 pipeline Lemma (5.1), frequency range below (9)                       |
| solve3(), min.beta() | Convention 2.2 | Appendix A (vertex enumeration)                                                                             |

## 9 Reproduction

Run `python3 fm_exact.py` (Python 3; `numpy/scipy` are used only for the final floating-point root-finding of  $\theta^{**}$ , which is verification of a closed-form calculus step and not part of the certification). Expected output: the exact table  $\zeta(\gamma) = 1/4, 3/16, 1/8, 1/16, 0$  at  $\gamma = 1, 21/20, 11/10, 23/20, 6/5$ , agreement with  $(6 - 5\gamma)/4$  throughout, the witness  $(\mu, \eta) = (11/20, 9/40)$  at  $\gamma = 11/10$ , the threshold  $\theta^{**} = 0.939305$ , and the margins  $+0.0136, +0.2028, +0.5478$  (in units of  $u$ ) at  $\vartheta = 0.94, 0.95, 0.97$ . Runtime is a few minutes.

## References

- [1] E. Fouvry and H. Iwaniec, *Exponential sums with monomials*, J. Number Theory **33** (1989), 311–333.

- [2] S. W. Graham and G. Kolesnik, *Van der Corput's Method of Exponential Sums*, London Math. Soc. Lecture Note Ser. **126**, Cambridge Univ. Press, 1991.
- [3] O. Robert and P. Sargos, *Three-dimensional exponential sums with monomials*, J. reine angew. Math. **591** (2006), 1–20.